

HHP ML Feature Correlation and Semi-Ablation Report

Prepared from the current code and output artifacts in /home/suramya/HHP-Prediction/OHC.

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Executive Summary

This report documents the correlation diagnostics, redundancy analysis, and semi-ablation experiments for the RTOFS-to-Argo residual-learning workflow used in the HHP project. The strongest robust feature choices are target-specific: `global_pruned` for TCHP and `drop_both_lat_interactions` for D26 under the locked blocked-forward protocol.

All added features are computed on the RTOFS side only. Argo is used only for collocation coordinates, observed truth, and residual targets.

1. Scope and Artifact Locations

To keep the PDF readable, paths are shown relative to the code root /home/suramya/HHP-Prediction/OHC.

- **Code root:** /home/suramya/HHP-Prediction/OHC
- **Collocation table:** output/ml_collocation/data/argo_rtofs_collocated_2024_2025.parquet
- **Global-physics table:** output/ml_collocation/data/argo_rtofs_collocated_2024_2025_physics.parquet
- **Profile-physics table:**
output/ml_collocation/data/argo_rtofs_collocated_2024_2025_profile_physics.parquet
- **Global feature builder:** build_rtofs_global_physics_features_2024_2025.py
- **Profile feature builder:** build_rtofs_profile_physics_features_2024_2025.py
- **Year ablation runner:** run_year_holdout_xgb_physics_semi_ablation.py
- **Locked ablation runner:** run_locked_xgb_physics_semi_ablation.py

The merged collocation table contains 41,347 rows across 90 dates. The locked protocol evaluates 8,366 out-of-fold validation rows across 57 dates.

2. Feature Construction

2.1 Interpolation rule

All RTOFS fields are sampled at the exact collocated Argo point using the same 8-neighbor native-grid interpolation rule. For non-exact points the implementation uses inverse-distance-squared weighting.

$$\begin{aligned} x_{\text{hat}} &= \sum_i (w_i x_i) / \sum_i w_i \\ w_i &= 1 / \max(d_i, 1e-6)^2 \end{aligned}$$

2.2 Which features are TEOS-10 and which are not

The steric-height features and Brunt-Vaisala frequency summaries are computed with TEOS-10 via gsw. SSH, MLT, SBLT, and the latitude interaction terms are RTOFS diagnostics or direct algebraic combinations.

Relative to the earlier tabular correction pipeline, the main additions in this report are: (i) model-side ocean-structure diagnostics sampled at the exact collocated point, (ii) steric-height summaries from the 3-D RTOFS profile, (iii) Brunt-Vaisala frequency summaries from the same profile, and (iv) semi-ablation experiments that explicitly prune correlated variants instead of only adding features.

2.3 Baseline features used across the ML pipeline

Before any added physics features were introduced, the baseline ML pipeline already used time, geography, collocation quality, and the raw collocated RTOFS state. These baseline inputs appear in the original tabular benchmark, the year-holdout runs, the locked protocol, and the later semi-ablation experiments.

Baseline feature	Short meaning	Why it is included
year	calendar year of the collocated row	time context
month_int	integer month 1-12	seasonality
lat	latitude of the collocated Argo/RTOFS point	geographic regime
lon	longitude of the collocated Argo/RTOFS point	geographic regime
abs_lat	absolute latitude $ \text{lat} $	latitude-band physics
nearest_rtofs_grid_distance_km	distance from target point to nearest native RTOFS cell	collocation quality
month_sin	$\sin(2 \pi \text{ month} / 12)$	cyclic seasonal encoding
month_cos	$\cos(2 \pi \text{ month} / 12)$	cyclic seasonal encoding
doy_sin	$\sin(2 \pi \text{ doy} / 366)$	cyclic annual encoding
doy_cos	$\cos(2 \pi \text{ doy} / 366)$	cyclic annual encoding
is_winter_jfm	1 for Jan-Feb-Mar rows	season flag
is_summer_jas	1 for Jul-Aug-Sep rows	season flag
is_other	1 outside JFM/JAS	season flag
model_interp_tchp_kj_per_cm2	raw collocated RTOFS TCHP at the Argo point	baseline model state
model_interp_d26_m	raw collocated RTOFS D26 at the Argo point	baseline model state

2.4 Named feature-set recipes used in the experiments

The experiment names in the ablation tables are shorthand for fixed feature recipes. This table gives a compact definition of the ones used in the current benchmarks.

Feature-set name	Summary
base	Baseline time, location, collocation-distance, and raw RTOFS TCHP/D26 features only.
base_plus_global_physics	Base plus the full global RTOFS-side physics family from the 2-D diagnostics and daily field products.
global_pruned	Base plus the pruned global physics subset: SSH, MLT, SBLT, temp-excess, D26-minus-MLT, D26-to-SBLT ratio, and three latitude interactions.
global_pruned_plus_profile_core	global_pruned plus steric_1000_ref2000, N2_max_upper200, and N2_mean_to_D26.
drop_temp_lat_interaction	global_pruned-style set with the temp-excess x $ \text{lat} $ interaction removed and profile-core terms added.
drop_ssh_lat_interaction	global_pruned-style set with the SSH x $ \text{lat} $ interaction removed and profile-core terms added.
drop_both_lat_interactions	D26-target set that removes both SSH x $ \text{lat} $ and temp-excess x $ \text{lat} $, keeps MLT x $ \text{lat} $, and retains profile-core terms.
surface_temp_swap	Same role as drop_temp_lat_interaction but swaps temp-excess for raw surface temperature.

2.4a Global/diagnostic added features

These are the globally available RTOFS-side features sampled from the daily field products and the 2-D diagnostic grid, then optionally combined algebraically with model D26 or latitude.

Global/diagnostic feature	Definition	Units	TEOS-10?
model_surface_temp_c	surface temp T_s	deg C	No
model_ssh_m	sea-surface height eta	m	No

Global/diagnostic feature	Definition	Units	TEOS-10?
model_mixed_layer_thickness_m	mixed-layer thickness MLT	m	No
model_surface_boundary_layer_thickness_m	surface-boundary-layer thickness SBLT	m	No
model_temp_excess_26c	T_s - 26	deg C	No
d26_minus_mlt_m	D26 - MLT	m	Derived
d26_minus_sblt_m	D26 - SBLT	m	Derived
d26_to_mlt_ratio	D26 / MLT	ratio	Derived
d26_to_sblt_ratio	D26 / SBLT	ratio	Derived
warm_layer_thickness_positive_m	max(D26 - MLT, 0)	m	Derived
model_ssh_x_abs_lat	eta * lat	m deg	No
model_mlt_x_abs_lat	MLT * lat	m deg	No
model_temp_excess_x_abs_lat	(T_s - 26) * lat	deg C deg	No

2.4b Profile added features

These profile features require the 3-D RTOFS archive column and TEOS-10 thermodynamic calculations. They are available on a smaller subset of rows because deep valid columns are needed.

Profile feature	Definition	Units	TEOS-10?
model_steric_0_1000_m	dyn_height(0/1000) / g	m	Yes: TEOS-10
model_steric_0_2000_m	dyn_height(0/2000) / g	m	Yes: TEOS-10
model_steric_1000_ref2000_m	steric(0/2000) - steric(0/1000)	m	Yes: TEOS-10
model_n2_mean_upper200_s2	mean(N^2; z <= 200 m)	s^-2	Yes: TEOS-10
model_n2_max_upper200_s2	max(N^2; z <= 200 m)	s^-2	Yes: TEOS-10
model_n2_mean_to_d26_s2	mean(N^2; z <= D26)	s^-2	Yes: TEOS-10
model_n2_max_to_d26_s2	max(N^2; z <= D26)	s^-2	Yes: TEOS-10

2.5 TEOS-10 mathematics used in the added profile features

Pressure and thermodynamic conversions are built from the collocated RTOFS profile column:

```
delta_z_k = delta_p_k / 9806
S_A       = SA_from_SP(S_P, p, lon, lat)
CT        = CT_from_t(S_A, t, p)
```

Dynamic height anomaly relative to reference pressure is converted to steric-height-style features in the current implementation as:

```
DeltaPhi(0/p_ref) = geo_strf_dyn_height(S_A, CT, p, p_ref)
H(0/p_ref)        = DeltaPhi(0/p_ref) / g
g                  = 9.81 m s^-2 # current code constant
```

Brunt-Vaisala frequency is computed with TEOS-10 and then summarized over depth windows:

```
N^2 = Nsquared(S_A, CT, p, lat)
mean(N^2 ; z <= 200 m)
max (N^2 ; z <= 200 m)
mean(N^2 ; z <= D26)
max (N^2 ; z <= D26)
```

2.6 Where SSH, MLT, and SBLT come from and why they were added

The SSH, mixed-layer thickness, and surface-boundary-layer thickness features are not computed from Argo and are not TEOS-10 outputs. They are sampled from the cached global RTOFS diagnostic NetCDF file `rtofs_glo_2ds_f006_diag.nc` at the exact collocated location using the same 8-neighbor interpolation rule. In code, these come from the variables `ssh`, `mixed_layer_thickness`, and `surface_boundary_layer_thickness` in `build_rtofs_global_physics_features_2024_2025.py`.

They were added because the original residual-learning baseline only saw location, seasonality, and model TCHP/D26 values. The new diagnostics give the model extra information about surface state and upper-ocean

structure without leaking Argo-derived physics into the inputs.

2.7 Feature availability

The correlation calculations use a complete-case subset. Availability therefore matters: the deepest steric and D26-limited N² features only exist on a small subset of rows.

Feature	Finite rows
model_surface_temp_c	41,224
model_ssh_m	41,224
model_mixed_layer_thickness_m	41,224
model_surface_boundary_layer_thickness_m	41,224
model_temp_excess_26c	41,224
d26_minus_mlt_m	12,814
d26_minus_sblt_m	12,814
d26_to_mlt_ratio	12,814
d26_to_sblt_ratio	12,814
warm_layer_thickness_positive_m	12,814
model_ssh_x_abs_lat	41,224
model_mlt_x_abs_lat	41,224
model_temp_excess_x_abs_lat	41,224
model_steric_0_1000_m	2,954
model_steric_0_2000_m	2,785
model_steric_1000_ref2000_m	2,785
model_n2_mean_upper200_s2	3,068
model_n2_max_upper200_s2	3,068
model_n2_mean_to_d26_s2	848
model_n2_max_to_d26_s2	848

3. Correlation and Redundancy Diagnostics

3.1 Pearson correlation coefficient used in the report

For two variables x and y over n complete-case rows, the Pearson coefficient is:

$$r_{xy} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 * \sum_i (y_i - \bar{y})^2}}$$

In this report, x is one added RTOFS-side feature and y is a residual target: either `delta_tchp_kj_per_cm2` or `delta_d26_m`. The current merged complete-case subset for the added-feature correlation test contains 795 rows.

3.2 Top absolute Pearson correlations with the targets

Feature	r with delta TCHP	r
d26_minus_sblt_m	0.199	0.199
d26_minus_mlt_m	0.179	0.179
warm_layer_thickness_positive_m	0.179	0.179
model_temp_excess_26c	0.152	0.152
model_surface_temp_c	0.152	0.152
model_n2_max_upper200_s2	-0.152	0.152
model_ssh_x_abs_lat	-0.132	0.132
model_mixed_layer_thickness_m	0.125	0.125
model_n2_mean_to_d26_s2	-0.114	0.114
model_steric_0_1000_m	0.113	0.113

Feature	r with delta D26	r
model_n2_max_upper200_s2	-0.198	0.198
model_mixed_layer_thickness_m	0.181	0.181
model_n2_mean_upper200_s2	-0.150	0.150
model_temp_excess_x_abs_lat	-0.143	0.143
model_n2_max_to_d26_s2	-0.137	0.137
model_n2_mean_to_d26_s2	-0.133	0.133
model_steric_0_1000_m	0.124	0.124
model_mlt_x_abs_lat	0.119	0.119
model_steric_0_2000_m	0.108	0.108
model_ssh_m	0.108	0.108

How to read these tables: the column `r with delta TCHP` means Pearson correlation between one added RTOFS-side feature and the residual label `delta_tchp_kj_per_cm2 = Argo TCHP - RTOFS TCHP`. Likewise, `r with delta D26` means correlation with `delta_d26_m = Argo D26 - RTOFS D26`. A larger absolute value means the feature tracks the signed correction target more strongly on the complete-case subset.

Immediate interpretation: for TCHP, the strongest linear relationships come from warm-layer geometry terms such as `d26_minus_sb1t_m` and `d26_minus_mlt_m`. For D26, the strongest linear relationships come more from stratification and structure terms, especially `model_n2_max_upper200_s2`, mixed-layer thickness, and the steric-height family.

3.3 High-redundancy feature pairs

Feature A	Feature B	r
model_surface_temp_c	model_temp_excess_26c	1.000
d26_minus_mlt_m	warm_layer_thickness_positive_m	0.996
model_steric_0_1000_m	model_steric_0_2000_m	0.974
model_surface_temp_c	model_temp_excess_x_abs_lat	0.958
model_temp_excess_26c	model_temp_excess_x_abs_lat	0.958
model_ssh_m	model_ssh_x_abs_lat	0.945
model_mixed_layer_thickness_m	model_mlt_x_abs_lat	0.933
model_steric_0_2000_m	model_steric_1000_ref2000_m	0.916
d26_minus_sb1t_m	warm_layer_thickness_positive_m	0.901
d26_minus_mlt_m	d26_minus_sb1t_m	0.900

The strongest duplicates are temperature-excess vs. surface temperature, warm-layer-thickness vs. D26-minus-MLT, and the multiple steric-height variants.

Interpretation: this table is not measuring skill against Argo directly. It is measuring redundancy between predictors. Very high `|r|` here means two features are carrying nearly the same information into the model, so keeping both can inflate feature importance instability without adding much new signal.

4. Semi-Ablation Results

4.1 Year-holdout semi-ablation (train 2024, test 2025)

This pass is useful for ranking feature-set candidates, but it is more optimistic than the locked protocol because it tests on only 29 dates in 2025.

In the year-holdout tables below, MAE and RMSE are errors against the Argo truth after applying the learned residual correction. `Gain vs raw` means the reduction in MAE relative to using raw collocated RTOFS alone.

TCHP model	MAE	RMSE	Gain vs raw
drop_temp_lat_interaction	11.504	16.619	4.666

TCHP model	MAE	RMSE	Gain vs raw
surface_temp_swap	11.504	16.619	4.666
drop_ssh_lat_interaction	11.541	16.659	4.628
global_pruned_plus_profile_alt	11.547	16.674	4.622
warm_layer_swap	11.557	16.707	4.612

D26 model	MAE	RMSE	Gain vs raw
drop_both_lat_interactions	11.277	15.115	3.546
drop_temp_lat_interaction	11.296	15.125	3.527
surface_temp_swap	11.296	15.125	3.527
drop_ssh_lat_interaction	11.310	15.177	3.512
global_pruned_plus_profile_core	11.319	15.168	3.503

Immediate interpretation: the year-holdout pass suggested a slightly different winner for each target. That was encouraging, but not yet enough to freeze the feature sets because the evaluation window was still comparatively narrow.

4.2 Locked blocked-forward semi-ablation

This is the stronger decision table because it aggregates 57 validation dates and forces each fold to predict forward in time with a one-date embargo.

Here, Corr. is the Pearson correlation between the corrected prediction and the Argo truth over the pooled out-of-fold rows. Bias is prediction minus truth, so a negative bias means the corrected model is still underestimating Argo on average.

TCHP model	MAE	RMSE	Bias	Corr.
global_pruned	11.639	16.385	-0.125	0.946
drop_temp_lat_interaction	11.657	16.443	0.289	0.946
surface_temp_swap	11.657	16.443	0.289	0.946
drop_ssh_lat_interaction	11.685	16.407	0.068	0.946
base	11.779	16.612	0.366	0.944
raw_rtofs	16.264	22.645	-8.969	0.915

D26 model	MAE	RMSE	Bias	Corr.
drop_both_lat_interactions	10.761	14.497	-0.411	0.930
drop_temp_lat_interaction	10.796	14.522	-0.321	0.930
global_pruned_plus_profile_core	10.836	14.569	-0.529	0.930
global_pruned	10.840	14.561	-0.579	0.930
base	10.979	14.832	-0.230	0.927
raw_rtofs	14.615	19.865	-7.276	0.888

Immediate interpretation: under the stricter locked protocol, D26 kept its semi-ablation winner (drop_both_lat_interactions), while TCHP reverted to the simpler and more robust global_pruned feature set. That reversal is precisely the kind of issue the locked protocol is meant to catch.

4.3 Seasonal locked results

TCHP season	Model	MAE	Bias	Corr.
other	global_pruned	11.869	0.581	0.940
other	drop_temp_lat_interaction	11.896	0.734	0.940
other	base	11.973	0.670	0.939
other	raw_rtofs	17.197	-10.122	0.905
summer_jas	global_pruned	11.811	-0.231	0.946
summer_jas	drop_temp_lat_interaction	11.828	0.230	0.946

TCHP season	Model	MAE	Bias	Corr.
summer_jas	base	11.926	0.450	0.945
summer_jas	raw_rtofs	16.514	-8.555	0.913
winter_jfm	global_pruned	10.691	-0.016	0.950
winter_jfm	drop_temp_lat_interaction	10.711	0.323	0.950
winter_jfm	base	10.969	-0.206	0.949
winter_jfm	raw_rtofs	14.550	-10.292	0.933

D26 season	Model	MAE	Bias	Corr.
other	drop_both_lat_interactions	10.802	-0.446	0.935
other	base	10.904	-0.776	0.933
other	global_pruned	10.993	-0.628	0.933
other	global_pruned_plus_profile_core	11.008	-0.632	0.933
other	raw_rtofs	14.879	-9.029	0.902
summer_jas	drop_both_lat_interactions	10.351	-0.514	0.934
summer_jas	global_pruned	10.420	-0.646	0.934
summer_jas	global_pruned_plus_profile_core	10.428	-0.680	0.934
summer_jas	base	10.590	-0.094	0.931
summer_jas	raw_rtofs	14.181	-6.408	0.888
winter_jfm	global_pruned_plus_profile_core	12.678	0.248	0.912
winter_jfm	drop_both_lat_interactions	12.690	0.102	0.912
winter_jfm	global_pruned	12.754	-0.228	0.910
winter_jfm	base	12.871	-0.572	0.909
winter_jfm	raw_rtofs	16.528	-10.424	0.883

Immediate interpretation: these grouped tables check whether the global winners are only winning because of one season. For TCHP, `global_pruned` stays competitive across winter, summer, and other months. For D26, `drop_both_lat_interactions` is strongest overall, especially in summer and the non-JFM/JAS subset, while the profile-core variant remains close in winter.

5. Interpretation

- The added RTOFS-only physics features are useful, but only after pruning correlated variants.
- The strongest exact duplicates are the expected construction pairs: `model_surface_temp_c` vs. `model_temp_excess_26c`, and `warm_layer_thickness_positive_m` vs. `d26_minus_mlt_m`.
- Profile-physics features help more for D26 than for TCHP. That is physically plausible because D26 is more directly tied to upper-ocean structure and stratification.
- The year-holdout pass suggested `drop_temp_lat_interaction` for TCHP and `drop_both_lat_interactions` for D26. After the stricter locked validation, the D26 winner held, but TCHP reverted to the simpler `global_pruned` feature set.
- Current recommendation: `global_pruned` for TCHP and `drop_both_lat_interactions` for D26.

6. File Reference Appendix

The table below lists the main output and source locations relative to `/home/suramya/HHP-Prediction/OHC`.

Artifact	Relative path
Collocation table	<code>output/ml_collocation/data/argo_rtofs_collocated_2024_2025.parquet</code>
Global-physics table	<code>output/ml_collocation/data/argo_rtofs_collocated_2024_2025_physics.parquet</code>
Profile-physics table	<code>output/ml_collocation/data/argo_rtofs_collocated_2024_2025_profile_physics.parquet</code>

Artifact	Relative path
Global feature builder	build_rtofs_global_physics_features_2024_2025.py
Profile feature builder	build_rtofs_profile_physics_features_2024_2025.py
Year ablation runner	run_year_holdout_xgb_physics_semi_ablation.py
Locked ablation runner	run_locked_xgb_physics_semi_ablation.py
Feature-feature corr matrix	output/ml_benchmarks/added_physics_feature_corr_pearson.csv
High-redundancy pairs	output/ml_benchmarks/added_physics_high_redundancy_pairs.csv
TCHP target corr	output/ml_benchmarks/added_physics_vs_delta_tchp_kj_per_cm2_pearson.csv
D26 target corr	output/ml_benchmarks/added_physics_vs_delta_d26_m_pearson.csv
Year ablation summary	output/ml_benchmarks/year_holdout_physics_semi_ablation_summary.csv
Locked OOF summary	output/ml_benchmarks/locked_physics_semi_ablation_oof_summary.csv
Locked grouped summary	output/ml_benchmarks/locked_physics_semi_ablation_grouped_all.csv